Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/23/2015
PI Name:	Zanello, Susana Ph.D.		
Project Title:	Brain Gene Expression Signatures From Cerebrospinal Fluid Exosome RNA Profiling		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	832-576-6059
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	Human Research Program Chief Scientist Office		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE (January 2021): PI now at KBR/NASA JSC as of December 2020. Previously at imec USA from June 2019-November 2020; NASA JSC (KBRwyle) from August 2017 until spring 2019. Prior to August 2017, PI was with Universities Space Research Association.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	10/01/2013	**End Date:**	10/01/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Allcorn, Aaron	**Contact Phone:**	281.244.8402
Contact Email:	aaron.j.allcorn@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: This project moved from an Internal Project to Grant NNX15AW48G starting 10/1/2015, per A. Allcorn/JSC HRP (Ed., 8/31/16) NOTE: End date changed to 10/11/2016 per information from A. Allcorn/JSC HRP (Ed., 5/16/16) NOTE: Extended to 2/26/2016 per R. Brady/JSC HRP (Ed., 3/3/15) NOTE: Extended to 9/30/2015 per PI (Ed., 2/27/15) NOTE: Extended to 1/31/2015 per PI (Ed., 7/15/14)		
Key Personnel Changes/Previous PI:	July 2015 report: Addition of new Co-investigator: Dr. Eric Bershad, Baylor College of Medicine		
COI Name (Institution):	Skog, Johann (Exosome Diagnostics Inc.) Carter, Bob M.D., Ph.D. (UC San Diego School of Medicine) Bershad, Eric (Baylor College of Medicine)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of the current priorities of the NASA Human Research Program is to characterize and quantify the risk of visual alterations and intracranial pressure (VIIP) associated with microgravity exposure during spaceflight. While the described VIIP syndrome focuses on ocular symptoms, spaceflight has been also linked to a number of other central nervous system (CNS)-related signs and symptoms. This proposal will evaluate a method of investigating brain physiology, molecular pathways and processes that may be altered by increased intracranial pressure (ICP). The proposed methodology consists in the identification of biomarkers by gene expression analysis of the transcriptome (mRNA) retrieved from exosomes in CSF and serum samples. Exosomes are microvesicles shed into all biofluids, including blood, urine, and cerebrospinal fluid, carrying a highly rich source of intact protein and RNA cargo, and thus constituting a representation of the physiology of inaccessible tissues like the brain. We will compare brain gene expression via exosome analysis in patients suffering from ICP elevation and normal ICP controls as a proof-of concept of this technique's value in finding measurable differences between the two groups. This will enrich knowledge towards the goal of characterizing and quantifying the risk. Once the methodology is validated, future work will aim at developing assays of diagnostic and prognostic value using the identified biomarkers. These assays may be applied to the astronaut population as well as patients suffering from ICP-related conditions on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has direct impact for the understanding of how increased ICP conditions may impact brain functioning and also for the understanding of the etiology of idiopathic intracranial hypertension. In addition, gene expression profiling from exosomes in the CSF is an emerging technology and its application on this research will help to develop its potential to be used for the characterization of CSF biomarkers in diseases of the central nervous systems and neurocognitive disorders.
Task Progress:	While the Visual Impairment and Intracranial pressure (VIIP) syndrome observations have focused on ocular symptoms, spaceflight has been also associated with a number of other performance and neurologic signs, such as headaches, cognitive changes, vertigo, nausea, sleep/circadian disruption, and mood alterations, which, albeit likely multifactorial, can also result from elevation of intracranial pressure (ICP). We therefore hypothesize that these various symptoms are caused by disturbances in the neurophysiology of the brain structures and are correlated with molecular markers in the cerebrospinal fluid (CSF) as indicators of neurophysiological changes. Exosomes are 30-200 nm microvesicles shed into all biofluids, including blood, urine, and CSF, carrying a highly rich source of intact protein and RNA cargo. Exosomes have been identified in human CSF, and their proteome and RNA pool is a potential new reservoir for biomarker discovery in neurological disorders. The purpose of this study is to investigate changes in brain gene expression via exosome analysis in patients suffering from ICP elevation of varied severity (idiopathic intracranial hypertension -IIH) as a first step toward obtaining evidence suggesting that cognitive function and ICP levels can be correlated with biomarkers in the CSF. Thus far, we have collected samples from 10 patients with elevated ICP due to trauma. The samples have been collected under IRB approved protocols at UC San Diego and shipped to the Exosome Diagnostics CLIA facility in Minneapolis. Analysis of these 10 patients, compared to 2 pooled control (normal ICP) samples of CSF from the ExosomeDX biorepository, has allowed us to conclude that exosomes from CSF can be efficiently used for RNA profiling. The isolated exosomes contained brain derived RNA transcripts as well as markers of inflammation. The expression trend of some of the inflammation markers examined suggest a concerted inflammatory response in response to the injuries. No correlation analysis with ICP was possible, since these samples corresponded to patients that underwent treatment for ICP reduction in the trauma unit. The next step will include samples from IIH patients with known ICP values. NOTE (Ed., 8/31/16): This project moved from an Internal Project to Grant NNX15AW48G starting 10/1/2015, per A. Allcorn/Johnson Space Center Human Research Program. Continues as part of "A Gene Expression and Histologic Approach to the Study of Cerebrospinal Fluid Production and Outflow in Hindlimb Suspended Rats," grant NNX15AW48G. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 09/04/2023)
Articles in Peer-reviewed Journals	Theriot C, Zanello S. "Molecular effects of spaceflight in the mouse eye after Shuttle Mission STS-135." Gravitational and Space Research. 2014 Aug;2(1):3-24. http://gravitationalandspacebiology.org/index.php/journal/article/view/643 ; accessed 7/24/15. , Aug-2014
Articles in Peer-reviewed Journals	Taibbi G, Cromwell RL, Zanello SB, Yarbough PO, Ploutz-Snyder RJ, Godley BF, Vizzeri G. "Ocular outcomes evaluation in a 14-day head-down bed rest study." Aviation, Space, and Environmental Medicine. 2014 Oct;85(10):983-92. http://dx.doi.org/10.3357/ASEM.4055.2014 ; PubMed PMID: 25245897; PubMed Central PMCID: PMC4240225 , Oct-2014
Journal/Magazine covers	Theriot C, Zanello S. "Cover in journal Gravitational and Space Research for article: 'Molecular effects of spaceflight in the mouse eye after Shuttle Mission STS-135.' " Gravitational and Space Research. 2014 Aug;2(1):3-24. http://gravitationalandspacebiology.org/index.php/journal/article/view/650/673 ; accessed 7/24/15. , Aug-2014
Significant Media Coverage	Goubet F. "Le corps human a l'epreuve du cosmos. Coverage for article on the one year mission on ISS and PI interview." Le Temps Newspaper, Geneva, Switzerland. March 27, 2015. http://www.letemps.ch/Page/Uuid/62a25f56-d3e5-11e4-9f2b-2a6998eb792a/Le_corps_humain_%C3%A0_l%C3%A9preuve_du_cosmos ; accessed 7/24/15., Mar-2015
Significant Media Coverage	Perrin S. "Living outside Earth is a huge challenge. Article in EPFL (Swiss Polytechnic Institute of Lausanne) website about PI Zanello's work." EPFL (Swiss Polytechnic Institute of Lausanne) website, May 2015: http://actu.epfl.ch/news/living-outside-earth-is-a-huge-challenge/ ; accessed 7/24/15., May-2015